SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

MineTwin

- **Problem Statement ID – SIH26009**
- **Problem Statement Title- AI/ML & Space Tech**

to Identify Manganese Reserves & Overcome Production Shortfalls

- **Theme- Space Technology**
- **PS Category- Software**
- **Team ID- 170155**
- **Team Name- RIT-Mine Twin**
- **App Name-MineTwin**



RIT-Mine Twin

Proposed Solution

SMART INDIA HACKATHON 2026

MineTwinAI: end-to-end intelligence for manganese exploration and planning

✦ Detailed explanation of the proposed solution

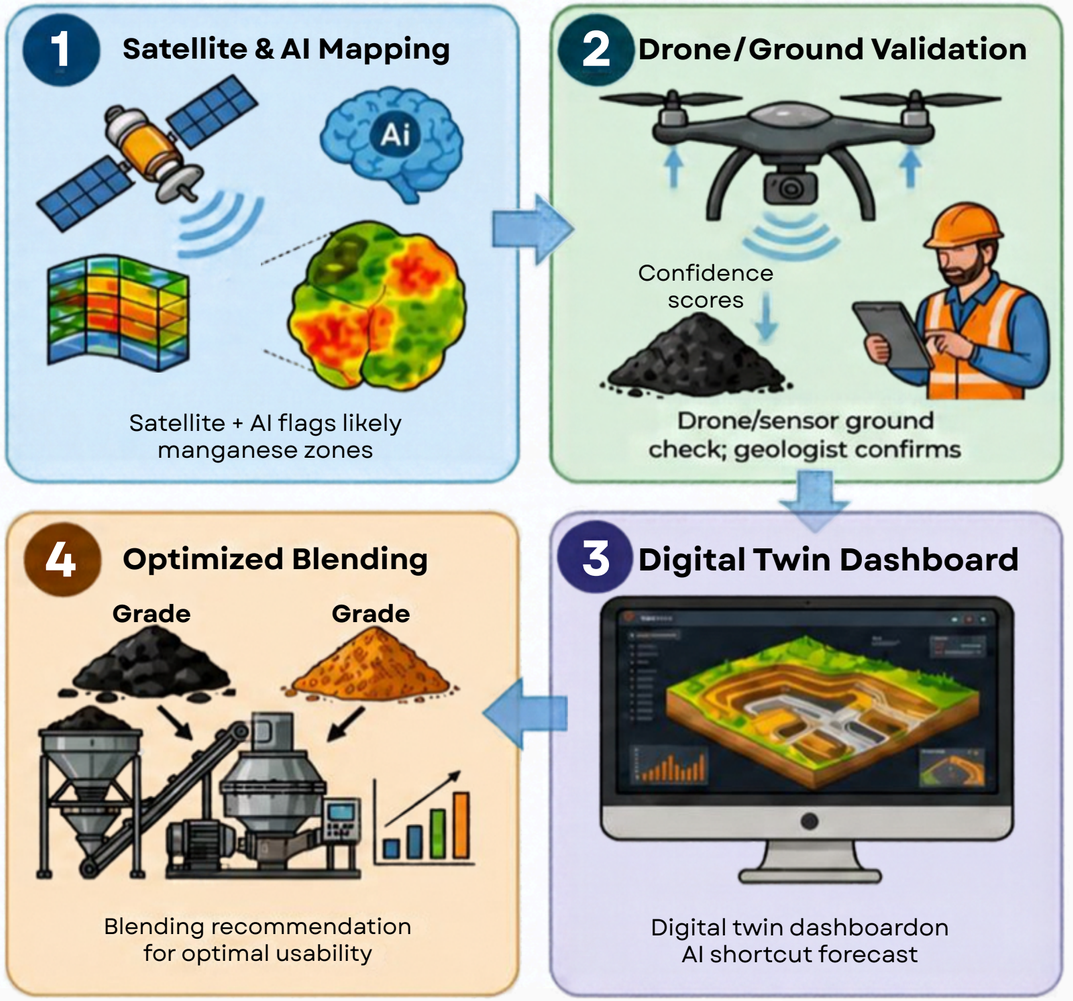
- Satellite + AI → Manganese Zones & Grade
- Drone/Sensor Validation → Geologist Approval
- Digital Twin → Shortfall Forecast + Blend Plan

✦ How it addresses the problem

- Lower Cost & Time → Smarter Exploration
- Low-Grade Stockpiles → Usable Blends → Fewer Imports
- Early Shortfall Alerts + Illegal Mining Detection

✦ Innovation and uniqueness of the solution

- Saptial ore reserve map→Ore grading by SWIR/NIR
- Grade-Aware Blending Optimizer → AI-prescriptive engine
- One Satellite Pipeline → Reserves + Illegal Mining
- 3D map→Block diagramatic representation→AI-copilot
- BlastSync schedule→Telementry →Preventive controller



RIT-Mine Twin

Technical Approach

SMART INDIA HACKATHON 2026

A five-step pipeline unifying satellite intelligence, sensing, and AI for smarter manganese planning

01 **Satellite + AI → Zone & Grade Mapping**
Analyze satellite imagery to map zones and estimate grade.

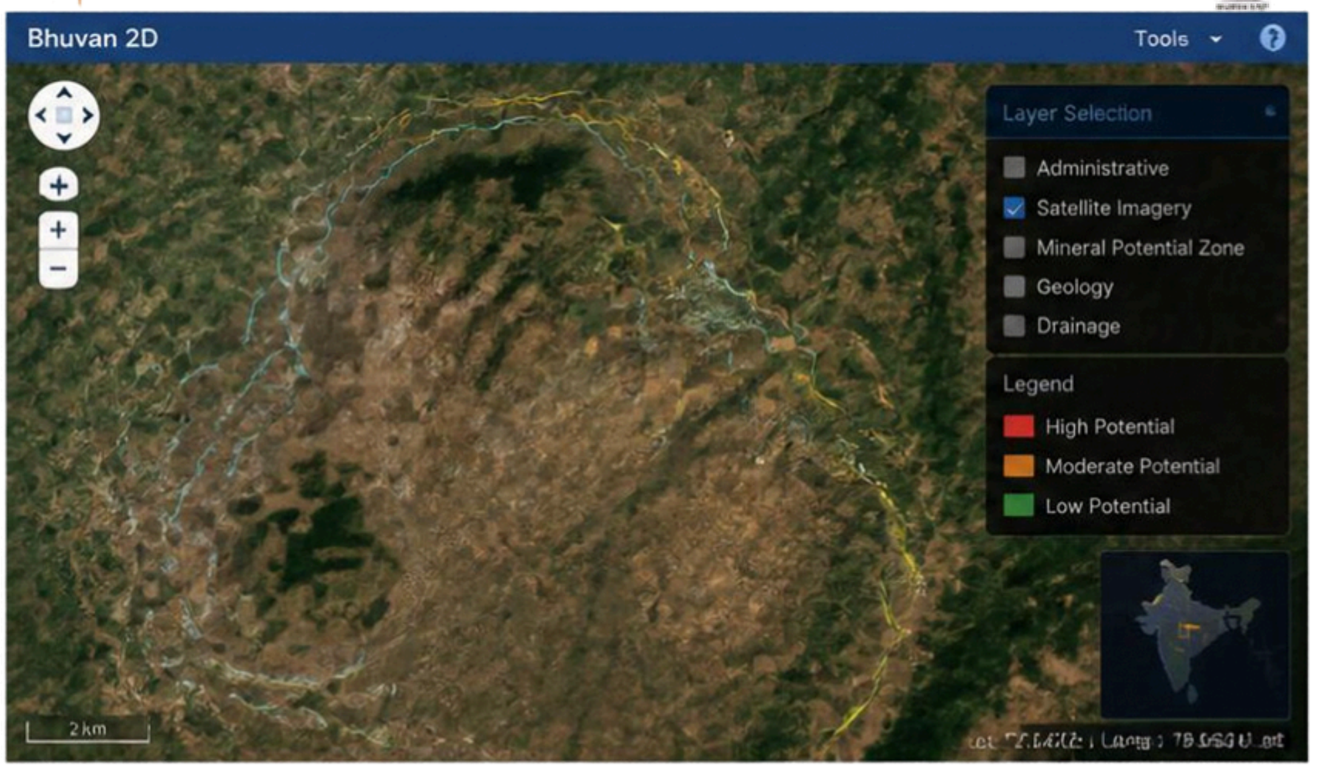
02 **Ground Sensors → Geologist Validation**
Collect field data and validate with geologist expertise.

03 **Digital Twin → Live Updates**
Create a digital twin that reflects real-time conditions.

04 **AI → Shortfall Prediction + Blend Ratio**
Predict shortfalls and optimize blend ratios with AI.

05 **Change Detection → Illegal Mining Alerts**
Monitor changes and alert on illegal mining activity.

Bhuvan
Indian Geo-Platform of ISRO



AI/ML
Python, CNN/Random Forest on ISRO Bhuvan & Sentinel-2 imagery

HARDWARE
ESP32 + spectral/soil/vibration sensors over WiFi/MQTT

BACKEND
JavaScript, Python, Flask,SciPy linear optimization

FRONTEND
HTML, CSS, React,Leaflet, Chart.js

SOURCE CODE
https://gITHUB.com/chilli360/MINETWIN.git

Feasibility and Viability

A practical path with clear risk controls



FEASIBILITY	RISKS	MITIGATIONS
Free satellite data availability (e.g., ISRO Bhuvan)	Spectral accuracy limitations	Human-in-loop AI with confidence scoring for trusted decisions
Sensor prototype cost under ₹5,000	Sparse data availability & connectivity challenges	Offline-first data capture with later synchronization
Phased validation: satellite screening to ground confirmation		Phased pilot deployment before scale-up to reduce risk
Modular architecture to reduce implementation risk		
Use of official satellite and mineral-resource data sources		

Demo Video Link: https://drive.google.com/drive/folders/1Ol3mLTRyZgQ9HsLnZxN5G7Ofk2JnHIr1

RIT-Mine Twin

Impact and Benefits

SMART INDIA HACKATHON 2026

Delivering measurable value across stakeholders and the mining ecosystem

STAKEHOLDER IMPACT

Mining Operators	Faster manganese zone discovery, optimized ore utilization, and reduced operational costs.
Government (GSI / IBM)	Automated illegal mining detection, real-time alerts, and efficient regulatory enforcement.
Steel & EV / Battery Industry	Reduced reliance on costly manganese imports via smart AI grade-blending.
Geologists & Researchers	High-confidence, satellite-linked mineral data and streamlined field validation.

BENEFITS BY CATEGORY

Economic	Lower overall mineral exploration costs. Maximized usability of low-grade ore stockpiles. Reduced national import dependency on manganese.
Social	Stable, sustainable livelihoods for regional mining communities. Transparent, safer, and well-regulated mining belts.
Environmental	Minimal land disruption through reduced blanket drilling. Mitigation of ecological damage caused by illegal mining.

MEASURABLE OUTCOMES ENABLED BY MINETWIN

30–40%
faster manganese-zone discovery vs. manual survey

15–20%
lower ore-export dependency via smart blending

Real-time
illegal-mining detection & enforcement alerts

Better prioritization of field surveys | Traceable evidence for review | More targeted exploration | Reduced dependence on blanket drilling

Research and References

Official sources informing our data, methods, and domain understanding

SOURCE	RELEVANCE	URL
Indian Bureau of Mines – Manganese Ore: Vision 2020 and Beyond	Official gov. mining vision	https://ibm.gov.in/
Ministry of Mines, GoI – Annual Reports on Manganese Production	Production data & trends	https://mines.gov.in/
ISRO Bhuvan/Resourcessource imagery portal	Satellite imagery source	https://bhuvan.nrsc.gov.in/
Geological Survey of India (GSI) – mineral resource maps	Mineral mapping & data	https://gsi.gov.in/
MOIL Limited – production & mine reports	Industry production reports	https://www.moil.nic.in/